

Enhancing the Historian.

Toward a Computational Critique of Historical Writing: Argumentation, Rhetorical Reflexivity, and Language Modelsⁱ

Abstract

This article presents an experimental system designed to assist historical writing, combining Retrieval-Augmented Generation (RAG) architectures and language models (LLMs). Unlike the dominant uses of these technologies in the humanities—which focus on documentary exploration or text generation—the proposed system aims to formalize certain recurring aspects of historical writing procedures by transforming them into analytical signals that historians can utilize in their own critical work.

The pipeline is based on a three-tiered structure of complementary analyses: the text's relationship to a bibliographic corpus, understood through a taxonomy of critical relationships derived from historiographical practices (confirmation, contradiction, nuance, problematization, displacement, particularization); the logical structure of the argument, modeled on the work of Toulmin and Walton's argumentative schemes; rhetorical strategies, analyzed within the framework of Perelman and Olbrechts-Tyteca's New Rhetoric, supplemented by Rhetorical Structure Theory for local discursive coherence. This combination produces a multi-level mapping of the argumentative forms at work in a text in the form of exploratory signals—not verdicts—that direct the historian's attention toward certain areas of the text that may warrant critical re-reading.

The architecture of the system is based on several guiding principles: computational frugality, user appropriation, code readability, and explicit attention to the sovereignty of research data. A distinction is made between a batch mode that processes an entire manuscript to produce orientation signals, and a mode of in-depth analysis on segments defined by the historian themselves. The stochastic nature of LLMs, which shapes the very style of programming, is embraced and documented rather than concealed.

Beyond its immediate applications, this work paves the way for a computational modeling of historical writing and helps define a research program at the intersection of digital humanities, argument analysis, and the epistemology of history. It proposes treating AI not as a substitute for the historian's judgment, but as an entity to be activated by encoding professional practices into it—a companion to the practice of history rather than a substitute.

Keywords

digital history · digital humanities · historical writing · large language models · retrieval-augmented generation · argument mining · Toulmin model · new rhetoric · Rhetorical Structure Theory · epistemological reflexivity · data sovereignty · semantic search · NLP pipeline · academic writing assistance · computational argumentation

Introduction

Recent developments in language models and semantic search techniques have profoundly transformed the conditions for accessing textual corpora. In the field of historical sciences, these tools now make it possible to query large documentary collections in natural language, to produce summaries quickly, and to assist in both the drafting and translation of texts. These transformations are part of a broader movement to reshape scholarly practices, initiated by computerization several decades ago and continuing today with the widespread adoption of artificial intelligence tools. They raise unprecedented questions about the nature of historical evidence and about the continuity—or discontinuity—between professional practices in the analog world and their digital rearticulations.

These transformations are accompanied, however, by a more subtle yet decisive shift: that of researchers' relationship to the digital tools they employ. When used as services provided by platforms—whose internal logic, formalization choices, and epistemic effects remain largely opaque—these systems tend to place the researcher in an essentially passive role, viewed as a user—or a customer. For the historical sciences, this shift raises a central issue: that of maintaining a reflexive mastery over the conditions of production of historical reasoning.

This article explicitly engages with this debate. It argues for the necessity—and the possibility—of an active and creative appropriation of digital tools in history, understood not as their mere use, but as their shaping, configuration, and transformation in the service of disciplinary practices. From this perspective, language models are not viewed as tools for delegating intellectual work, but as objects to be configured and activated.

It is within this framework that an experimental tool for assisting historical writing is proposed, one that can also be deployed in an educational context. This tool is based on Retrieval-Augmented Generation (RAG) architectures and language models, not to generate text or optimize access to information, but to analyze reasoning as it is being developed. This prototype aims to validate the feasibility of a methodological proposal that can also be understood as the outline of a research program, intended to build upon research developments over the past decades. Technical details, complete scripts, variability tests, and output examples are available in the associated GitHub repositoryⁱⁱ. This paper focuses on the conceptual principles, general architecture, and epistemological implications.

A first generation of tools resulting from these developments focused primarily on document exploration and content generation. RAG-type architectures, which combine vector search with language model generation (Lewis et al., 2020), have notably made it possible to query researcher-controlled documentary corpora using natural language. These uses are now well documented, even if their effects on historical practices remain largely understudied in a systematic manner.

However, these approaches largely overlook a central dimension of historical practice: the construction, validation, and testing of reasoning, as well as its confrontation with existing historiography. Writing history does not consist of proposing a narrative synthesis; it rests on a series of critical operations that articulate the relationship to sources and historiography, the

logical structuring of arguments, and the discursive strategies through which these arguments are rendered acceptable within a given framework.

Every historian is familiar, in this regard, with that moment when they gauge the gap between what they know they have read and what they should have read: the hundreds, sometimes thousands, of publications related to the subject they are addressing, in which someone may have already formulated the argument they believe they are developing, or, on the contrary, demonstrated its limitations. It is from this concern—both documentary and argumentative—that the framework presented here was partly born.

The integration of language models into historical practices also poses a specific epistemological problem. While these tools facilitate access to sources and the production of text, they also tend to obscure the processes through which historical text comes into being: the selection of evidence, the sequencing of arguments, the construction of generalizations, and positioning within historiographical debates. The risk is not only that of factual errors—LLM hallucinations are now well documented (Maynez et al., 2020)—but also that of a loss of control over the methods of validating historical arguments, in other words, a weakening of the reflexivity that underpins the scientific nature of historical work.

This study proposes to shift the focus of these applications. Rather than treating language models as tools for content generation or documentary exploration, it employs them as instruments for analysis and for testing reasoning—a mechanism to be activated by encoding professional practices into them. The framework thus combines a logic of assisted documentary exploration with a logic of critical analysis, comparable to the process of re-reading practiced by every historian.

The proposed pipeline is based on a three-tiered structure of complementary analyses: the text's relationship to a bibliographic corpus, understood through a taxonomy of critical relationships; the logical structure of the argument, modeled on the work of Toulmin and Walton's argumentative schemas; and rhetorical strategies, analyzed within the framework of New Rhetoric and *Rhetorical Structure Theory*. This combination enables the production of a multi-level mapping of the argumentative forms at work in a text, in the form of exploratory signals guiding critical re-reading, rather than automated diagnoses.

This paper first presents a state-of-the-art review and situates the framework within the existing literature, before describing the architecture of the pipeline and its design principles, and then outlining the three proposed levels of analysis and their interrelationships. It concludes with a discussion of possible future extensions of this work, envisioned as a contribution to a digital history grounded in the critical and reflexive appropriation of computational frameworks.

State of the art

Recent research has led to the development of methods for analyzing, exploring, and structuring large-scale text corpora. Early approaches, based on lexical statistics and lemmatization, made it possible to identify vocabulary patterns, co-occurrences, and diachronic changes that cannot be detected through ordinary reading of large volumes (Sinclair & Rockwell, 2016; McCarty, 2005). The procedures developed at that time, often encoded in standalone software tools (Lexico3 (Lamalle Martinez 2001), Txm (Heiden, Magué, Pincemin, 2010)), are part of the traditions of corpus linguistics and French-speaking textometrics, which very early on integrated lexicometry and hermeneutic interpretation. Grouped under the successive labels of *text mining*, *corpus linguistics* applied to the humanities, and then *distant reading* (Moretti, 2013), they share the common approach of treating text as an object whose statistical properties emerge at the corpus level rather than at the level of the document or excerpt. The value of these approaches has never been universally accepted; indeed, they have drawn sharp criticism, the most insightful of which has come from authors who emphasize the importance of maintaining a back-and-forth between *distant* and *close reading* and of adopting a methodological agnosticism that allows for the integration of tools from distinct fields (Underwood, 2019). Historians have adopted these methods to varying degrees. Pioneering work in quantitative history had already shown, as early as the 1960s and 1970s, that the serial processing of documentary sources could reveal structures invisible to cursory reading (Fogel & Engerman, 1974; Furet, 1982). This seminal moment provides a direct methodological lineage to current computational approaches, shifting the focus from the study of singular cases to observable statistical patterns and favoring changes of scale. The massive digitization of primary and secondary sources, which accelerated in the 2000s and 2010s, opened up new perspectives by making corpora of unprecedented scale accessible—digitized press, administrative archives, correspondence—to which computational approaches could be applied, altering both the conditions of circulation within vast documentary corpora (Burrow 2010) and the conditions of their analysis (Putnam, 2016). These corpora remain, however, deeply shaped by the documentary and technical production chains that governed their digitization, which implies treating digital availability as a social and technical fact that must be taken into account in the analysis (Drucker, 2021).

The turn of the 2010s saw the emergence of methods based on dense vector representations of language, first with Word2Vec (Mikolov et al., 2013) and then with *transformer architectures* (Vaswani et al., 2017). The Transformer architecture made it possible to train very large-scale models capable of capturing long-range dependencies in texts, directly paving the way for contemporary large language models. These representations allow for the capture of semantic relationships that classical statistical approaches could not grasp—synonymy, analogy, thematic proximity. They have profoundly transformed the possibilities for querying very large corpora, enabling the mapping of the semantic evolution of historical concepts over the long term (Hamilton et al., 2016) and the identification of recurring lexical structures at the scale of massive corpora to study variations in their distribution across different periods (Underwood, 2019).

The introduction of large language models, starting with GPT-3 (Brown et al., 2020) and its successors, marked another breakthrough. These models demonstrate capabilities for understanding and generating natural language that now make it possible to query corpora in natural language, summarize documents, classify texts according to user-defined categories, and produce unsupervised analyses. These tasks correspond to intellectual operations traditionally central to the work of historians—such as note-taking and pattern recognition—which are now partially delegated to a computational system. Combined with retrieval-

augmented *generation* (RAG) architectures (- RAG - Lewis et al., 2020), they enable the generative power of LLMs to be connected to specific document corpora that the model did not encounter during training, thereby paving the way for systems capable of answering questions in natural language by drawing on a library of sources controlled by the researcher. This LLM-RAG coupling is now one of the main paradigms of document-centric AI. Generation is no longer conceived as autonomous, but as an interface for reformulating and linking curated sources.

These developments are accompanied by critical reflection on the epistemic limits of these applications: the ability of LLMs to produce factually incorrect statements presented with confidence (Maynez et al., 2020), dependence on biases encoded in training data (Bender et al., 2021, 613), and difficulty distinguishing what the model has learned from what it reconstructs (Bommasani et al., 2021). These limitations are not merely technical flaws requiring engineering solutions; they stem from the probabilistic nature of these models and, in the context of their use in historical research, from the way they reconstruct the past based on contemporary textual distributions.

Questions regarding the use of these tools in a historical context leave some gaps. They are generally viewed as a means of querying external corpora, rather than as tools employed by historians in their own writing—for example, in the analysis and revision of a text as it is being written. In short, the historian, a passive operator at a terminal, fades into the background in favor of the completeness of the corpora, the granularity of the data, and the efficiency of the algorithms. It is possible, however, to bring back into focus the gestures, strategies, and dialogues that shape the historian’s work and determine results and conclusions.

Research in computational *argument mining* has shown that the automatic extraction of argumentative structures from academic texts is possible, particularly by applying Toulmin’s framework (Lawrence & Reed, 2020; Stab & Gurevych, 2017), but this research primarily targets deliberative or scientific texts, whose argumentative structure is fairly explicit, rather than texts whose argumentative structure is often implicit, polyphonic, and constructed over the extended duration of a chapter or a book. This makes the use of their modeling rare in the case of historical texts characterized by the polyphony of cited voices and the weaving of arguments over the long term. Perelman and Olbrechts-Tyteca’s *New Rhetoric* (1958) is virtually absent from *the* field of computational *argument mining*, despite its particular relevance to academic texts in the humanities. This absence has been noted in recent reviews of *argument mining* (Lippi & Torroni, 2016), though it has not yet led to a genuine integration program. It is in this space—between existing corpus query tools, computational argument analysis tools, and the tradition of instrumented history—that the present framework aims to position itself.

Architectures

The proposed system builds upon *Retrieval-Augmented Generation* (RAG)-type pipelines but shifts their objectives. While these architectures are generally designed to optimize access to extensive documentation in order to produce relevant responses, the pipeline presented here aims to extend these operations through a series of analyses focused on evaluating the reasoning and arguments deployed, positing that the operations of documentary search and generation are themselves already acts of argument construction.

All components of the pipeline were coded in Python as modular scripts, following an architecture that adheres to several guiding principles, the reconciliation of which constitutes a classic constrained optimization challenge (Wieringa, 2014).

Computational frugality is a primary imperative. It entails both minimizing calls to external services and conserving computational power whenever possible. This falls within the framework of digital sobriety, which Bordage (2019) has shown involves not only technical choices but also a design approach aimed at reducing the systemic footprint of software. In the field of computational social sciences and humanities, where resources are often limited, this principle determines the sustainability of the tools produced.

The concern for fostering user adoption constitutes a second guiding principle. This is reflected in the systematic grouping of parameters and settings at the beginning of the script. This facilitates the location and modification of control variables without the need to read the entire code. It also involves the provision of fine-tuning options, which are actionable because the pipeline is accompanied by explanations of their meaning and effect. This approach aligns with research on *end-user programming* and digital literacy (Burnett & Myers, 2014): technical mastery, which is never guaranteed, requires not only access to tools but also the ability to understand their configuration mechanisms.

We also aim to foster an understanding of the mechanisms at work, and even to facilitate learning—a goal reflected in the extensive documentation, which some may find verbose. It aims to be as precise and readable as possible, translating the programming logic underlying the use of tools by users with limited computer literacy into natural language, while also demonstrating that code is a text that expresses an attitude, a position, and sometimes a thesis. This pedagogical concern is part of the long tradition of work on *literate programming* initiated by Knuth (1984). A program must first be conceived as a text intended to be read and understood by humans, with the machine serving only as a secondary recipient. More recent formulations of these principles, often emerging from the digital humanities, use different terminology but align with these recommendations by emphasizing the need to make computational tools intelligible to non-computer science researchers (Ramsay & Rockwell, 2012).

Reconciling these different imperatives is difficult. A script that is extensively commented and structured for ease of use is longer and potentially slower than a script optimized solely for performance. Modularity facilitates understanding and reuse, but can introduce additional costs related to function calls and dependency management. These tensions have been accepted rather than resolved, by systematically prioritizing readability and learnability over raw performance.

The choices made can be read, understood, and critiqued within this framework; Python, as the chosen programming language, provides an example. This language is not the most computationally efficient—compiled languages such as C++ or Rust offer significantly superior performance (Pereira et al., 2017). However, it is currently the most widely accessible language in academic communities (Stack Overflow Developer Survey, 2023), and the training corpora for large language models include very large corpora of Python code: it is therefore particularly well supported by LLM-type assistants. This helps create the conditions for user adoption by allowing users to easily obtain automatic code redocumentation or adaptation advice. In the case of a script pipeline without a front end, this is likely a necessity. This benefit comes at a cost: it indirectly contributes to the use of particularly energy-intensive devices, a trade-off accepted on the assumption that the spread of digital literacy is itself a prerequisite for energy efficiency. Let us add, however, in this case—and this may be a decisive argument—that Python is used here only as a lightweight orchestrator: the bulk of the computation time (approximately 95%) results from the use of native libraries or LLM calls, most of which are intended to be executed locally.

Orchestrating a probabilistic agent

This pipeline, which is a prototype and could even be described as an encoded historical narrative, does not consist of a set of deterministic scripts. It serves as an orchestrator of language models. The behavior of these models is fundamentally probabilistic. This difference in nature has implications for how the code is written and how its outputs should be interpreted.

A classical algorithm allows one to control a flow of instructions whose result is predictable based on the inputs. When orchestrating an LLM, one negotiates with an agent that can interpret, rephrase, or partially ignore—and what “partially” encompasses can vary considerably—the given instructions, even those formulated most strictly. This is what Licklider (1960) had already identified, in a different technical context, as the fundamental difference between automating a task and collaborating with a system capable of initiative. Here, the collaboration takes place with an agent whose behavior is stochastic by nature.

This property has several implications. The regular expressions used to extract results from LLM responses are deliberately tolerant rather than strict. There are not many other options besides looking for a probable pattern rather than a guaranteed format, accepting variations in case, punctuation, positions, or phrasing that the model produces, despite the persistence and ingenuity deployed in writing the prompts. Parsing functions systematically include fallback mechanisms for cases where the LLM has produced malformed JSON, omitted a requested section, or restructured its response in an unexpected way. In fact, it sometimes displays surprising creativity, though rarely welcome. Intermediate backup and resume-on-interruption systems are integrated into the batch processing scripts, which can analyze several hundred paragraphs in a single run. The models’ temperature is kept low, between 0.05 and 0.1 depending on the scripts, to reduce output variability.

Debugging these scripts, moreover, is of a particular nature. We are not looking for a logical bug—or, more precisely, not only that—but rather a deviation in probabilistic behavior. This requires testing across various configurations. This stochastic dimension is an accepted limitation of the system, explicitly documented in the code and in the outputs, rather than an imperfection to be corrected.

The choice of language model: an epistemic variable

The pipeline relies on language models to produce argumentative and rhetorical analyses. The choice of these models is not neutral. The quality of the analyses produced depends directly on these choices. In addition to the raw power of the models, several factors influence the nature and quality of the responses.

The first is the density of representation of the theoretical frameworks used in the training data. Those utilized by the pipeline—Toulmin (1958), Perelman & Olbrechts-Tyteca (1958), Mann & Thompson (1988)—are theories present in the academic literature in philosophy, linguistics, and rhetoric; their long-standing presence and very classicism serve as a guarantee of this. The effectiveness of prompts utilizing these frameworks depends on a condition of possibility rarely stated explicitly: that they are already represented, with sufficient density, in the training corpora of the models used. This condition is not trivial; verifying that it is met is difficult, if not impossible, but it is likely true for the theories used here. Toulmin and Perelman are classic references in analytic philosophy and rhetoric, extensively discussed in academic literature in philosophy, linguistics, communication sciences, and in textbooks on argumentation—all of

which are overrepresented in the training data of major language models. A model can thus, without ever having been fine-tuned on a specific argumentative task, identify the components of a Toulmin schema or characterize a Perelmanian association technique, because texts describing and applying these frameworks constitute a significant portion of its training corpus.

This hypothesis finds empirical support in recent work. Chen et al. (2024), evaluating the performance of several LLMs on fourteen *argument mining* datasets in a zero-shot setting, show that these models achieve results comparable to, or even superior to, previous supervised approaches on the majority of argumentative tasks. Castagna et al. (2024) demonstrate that structuring prompts based on the critical questions associated with Toulmin’s model significantly improves the reasoning capabilities of LLMs. These zero-shot or quasi-zero-shot performances suggest that the models’ familiarity with classical theoretical frameworks of argumentation is not an uncontrolled assumption, but a documentable property of their training corpus and a necessary condition for certain developments.

This dependence on training data, however, constitutes a limitation that must be explicitly acknowledged. As Bender et al. (2021) point out, the analysis produced by an LLM reflects not only the theoretical framework invoked in the prompt, but also the way in which that framework is represented—and potentially distorted—in the texts from which the model learned. As a result, the framework proposed here does not produce a Toulmin or Perelman analysis in the strict sense, but rather an analysis informed by the representation of these theories offered by the training corpora. This distinction is fundamental to the pipeline’s epistemology: it invites us to treat its outputs not as mechanical applications of a theoretical framework, but as probabilistic signals guiding attention—a point the system explicitly acknowledges.

The ability to follow constrained output formats is, moreover, very unevenly distributed. The pipeline requires models to produce responses structured according to strict formats—numerical scores, labeled Toulmin components, RST trees in JSON. This capability, developed through reinforcement alignment techniques (Ouyang et al., 2022), varies significantly across models and directly determines the reliability of automatic result extraction, even if the regexes used are the most tolerant ever devised.

In the version documented here, the pipeline uses gpt-4.1-mini (OpenAI) for API-based analyses and qwen2.5:14b (Alibaba Cloud, via Ollama) for local analyses. These choices are largely arbitrary. For API-based use cases, numerous alternatives exist; as far as we can accurately determine, academic French is well-represented in the training corpora of most major models. The generic nature of the oldest widely available LLM alone justified this choice during the design phase. For local use cases, several *open-source* models offer credible alternatives with a comparable memory footprint. We simply chose the most powerful model among those that could be deployed on the machine used for testing when the tests were conducted. This also corresponds to what can be used on a laptop with 16 GB of RAM—a configuration that is not uncommon today—as well as the minimum size required for the pipeline to produce relevant results, since smaller models proved unsuitable.

The model’s Chinese origin warrants mention in a European research context. Chinese models respect data ownership no more than others do—that is, not very much. However, local use resolves issues of sovereignty as well as intellectual property: no data is transmitted to external servers. Regardless of cost considerations, this naturally argues in favor of using a local architecture, which also has the advantage of relying on energy infrastructures that heavily

favor renewable and nuclear sources while avoiding the use of an external API that encourages intensive or even compulsive usage.

The LLM landscape is evolving at a pace that renders any model choice obsolete in the short term. This is a limitation we accept here and have encoded into the scripts, which allow for relatively easy model switching. Inter-model variability—and even intra-model variability depending on the temperature and context—is a source of uncertainty that the pipeline attempts to mitigate through a low temperature and constrained prompt formats, but cannot eliminate. This model dependency is accentuated here by the fact that the system deliberately relies on the nature of the training corpora provided to large models. This variability is analogous, in a sense, to inter-rater variability in manual annotation protocols in corpus linguistics (Krippendorff, 2004): it must be documented and acknowledged rather than concealed. It can also, at least to some extent, be measured—something we have not done at this stage, since this pipeline was not intended to produce stable or reliable metrics but rather to generate user-actionable signals in a business context where expert judgment is, by the very architecture, entrusted to the user.

Data Sovereignty and Modular Architecture

The issue of research data sovereignty is rarely addressed explicitly in work on the uses of LLMs in the digital humanities, even though it directly determines the institutional adoption of these tools. Within a manuscript analysis pipeline, two types of data circulate separately: bibliographic corpus data, on the one hand, and data from the manuscript being written, on the other.

The RAG architecture is based on a functional separation between two independent components: an *embedding* model that produces vector representations of texts and enables semantic similarity search, and a language model that receives the selected passages and produces the analysis (Lewis et al., 2020). The generation model never sees the vectors; it receives the text passages that the vector search has identified as relevant. This separation is an architectural property that allows for complete modularity: the choice of vectorization model and the choice of generation model are independent of one another and can be freely combined. Recent developments suggest that this independence has its limits and is not guaranteed in the long term, but it remains the case as I write these lines and in the context of a system of the type discussed here.

This independence has direct implications for data sovereignty. Locally available multilingual *embedding* models—notably the multilingual-e5 family (Wang et al., 2024) and the *SentenceTransformers* models (Reimers & Gurevych, 2019)—produce vector representations of comparable quality to proprietary models for this type of task, at zero cost and without data exposure. The FAISS vector index (Johnson et al., 2021) is built and queried locally. The bibliographic corpus therefore never passes through external servers if these local components are chosen.

In this regard, the pipeline distinguishes three stages with very different sovereignty properties, corresponding to three choices the user must make.

The first step is the construction of the vector corpus. The documents in the bibliographic corpus are divided into segments, encoded into vectors by an *embedding* model, and indexed in a FAISS database. This step is entirely local if a local *embedding* model is used: no data is transmitted externally, and the vector index resides on the researcher's machine.

The second step is the manuscript-corpus comparison. For each paragraph of the manuscript, FAISS selects the most semantically similar passages from the corpus. The LLM receives the paragraph and the selected passages and classifies their relationships according to the taxonomy encoded in the scripts. If an external LLM is used at this stage, the entire manuscript is transmitted to third-party servers paragraph by paragraph through successive calls, along with a significant portion of the documentary corpus used. Even if this is done in the form of short segments, it amounts to providing a commercial operator with material—some of which may be copyrighted—free of charge. Commercial companies offering access to an LLM certainly generally provide very virtuous clauses regarding data ownership, but it is quite doubtful that they are respected. Using a local model at this stage, even if it generally comes at the cost of lower analysis quality, allows one to control the exposure of one’s manuscript. In the case of historical works, one might consider that sovereignty concerns are minimal and that no one will lose a war or a market due to the premature release of a manuscript; however, intellectual property concerns remain, as do those related to the author’s personal safety under a surveillance regime.

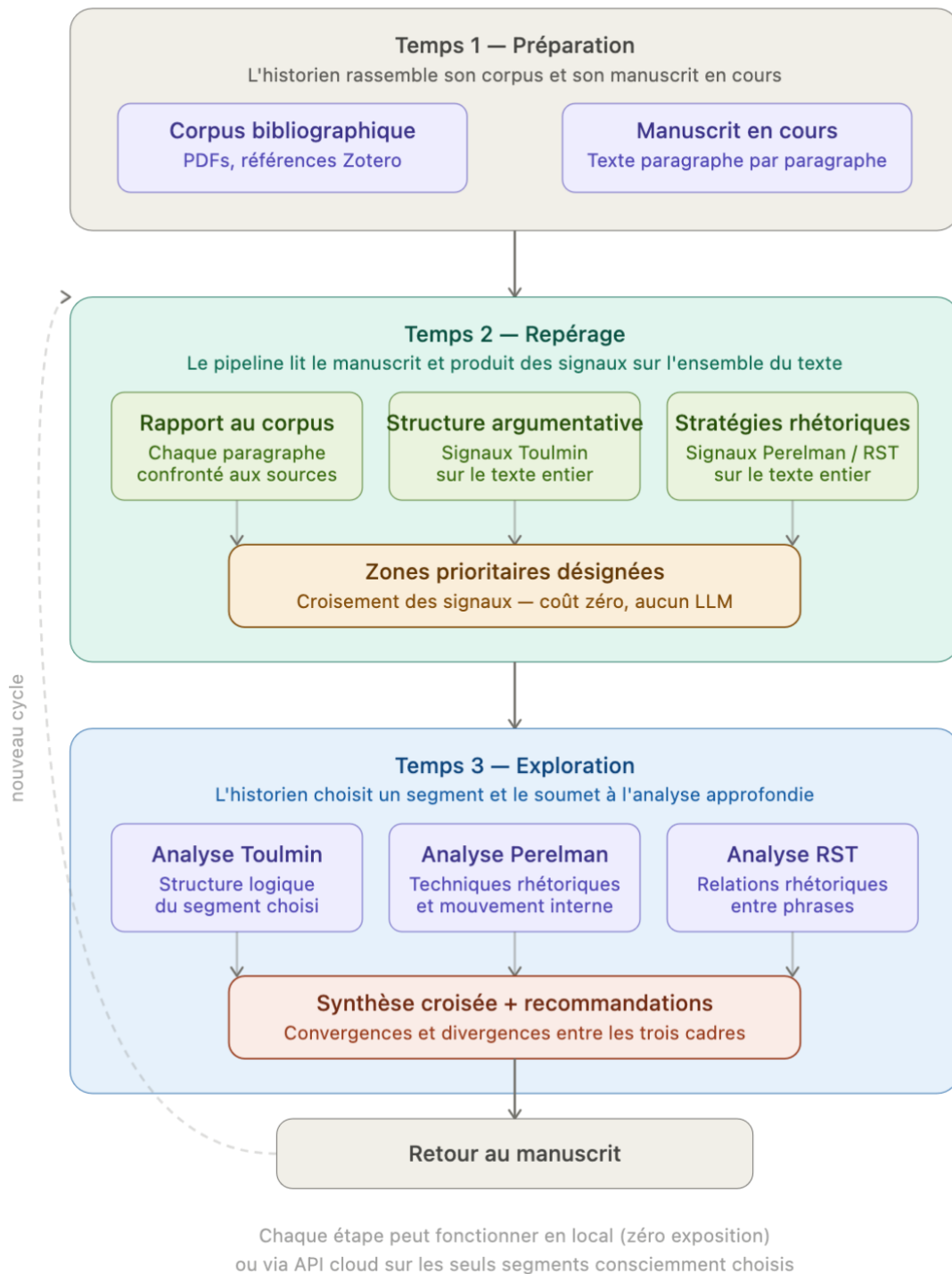
The third step is in-depth segment-by-segment analysis. The historian personally identifies the passages of their text to be subjected to a Toulmin, Perelman, or RST analysis. If the decision is made to use an external model, only these segments are sent to the external LLM. This constitutes a consented and controlled exposure, as opposed to the systematic exposure of the previous step.

This three-part division is more practical here than the generic local/cloud distinction, because it corresponds to the choices the user must make at each stage of processing, thereby highlighting their status as a situated, expert, and informed operator. A researcher working on sensitive archives or hypotheses whose primacy is at stake has legitimate reasons to keep their entire pipeline on-premises. A researcher whose corpus consists of published printed sources and whose manuscript does not contain sensitive unpublished results may use an external LLM to benefit from the most powerful models on the segments they deem a priority. The modularity of the architecture makes this decision explicit and reversible rather than imposing it by default, which makes sense within the framework of a system that can be used in multiple national contexts and during a period in which the standards, guidelines, and recommendations of the institutions to which it is affiliated may change, often evolving in a stochastic manner. This choice is also in line with recent thinking on *privacy-preserving AI* in institutional contexts (Mireshghallah et al., 2020).

The stages of the pipeline are described in detail in the following section. Suffice it to say that the architecture distinguishes between two modes of operation with different properties: a batch mode that processes an entire manuscript paragraph by paragraph, generating orientation signals, and a mode of in-depth analysis applied to segments defined by the historian themselves, which applies theoretical frameworks to coherent textual units. This distinction, which addresses the limitations identified in the literature on automatic *argument mining* (Peldszus & Stede, 2013), structures the entire system and determines the interpretation and use of the results produced.

What the pipeline does

The historian's journey



Historiographical Anchors

The first level of analysis proposed by the pipeline focuses on the relationship between the manuscript and a corpus of texts. In the context of the development of this paper and this system, the manuscript pertained to the history of extradition and its media representations in

the 19th century. Two accompanying text corpora were used: one contained historical legal treatises and articles, which addressed copyright and intellectual property challenges during the development phase when interacting with an external entity; the other comprised a set of academic articles related to the subject under study.

In traditional uses of NLP architectures, passages selected from the corpus are primarily evaluated in terms of relevance. They are intended to provide an adequate context for generating a response. We shift this approach, considering that the relationship between a statement and the materials associated with it is not merely a matter of relevance, but involves a broader range of critical relationships that characterize how the author of a text aligns with or opposes—or does not—the texts produced by their predecessors or contemporaries. This is not a dialogue with a corpus of sources, even though this framework is certainly transferable to other contexts, and even though ancient legal treatises can be considered printed sources. The interaction takes place with a corpus of texts containing statements considered a priori relevant and legitimate, because they correspond to the historiographical horizon defined by the historian themselves, without requiring the formalization of a historical critical operation.

This system offers the classic functions of a Natural Language Processing (NLP) system. It allows the constructed corpus to be used as a knowledge base within which the most relevant segments regarding the posed question are identified, with the possibility of obtaining a summary of these in natural language. The selected segments are cited in the body of the text with a reference to their bibliographic details, ensuring the traceability of the proposed statements and citations and allowing for direct verification. The system can be connected to a Zotero collection, a tool widely used in the historical community.

A simple example will illustrate this. In response to the question “Write a paragraph explaining why a historian should prefer Gallicagram to Google Ngram,” the pipeline returns a written passage that identifies twelve relevant segments in the corpus (the number of relevant segments selected is configurable), synthesizes them into a structured argument, and, where possible, accompanies each statement with a reference in the form [*DeCoursonAzoulay2021.pdf*, p. 4] or [*CoursonAzoulay2023.pdf*, p. 7]. The historian receives a writing proposal grounded in the texts they have gathered themselves, along with references allowing them to trace the source to verify the exact context.

The pipeline also makes it possible to link a segment of the manuscript being written—by default a paragraph, based on the assumption that the paragraph corresponds to a unit of meaning—to the bibliographic corpus, by identifying the segments semantically closest to it and suggesting their integration into the existing argument. When a passage from the corpus is deemed relevant, the system identifies the most useful citation—that is, the closest one—explains its argumentative function within the context of the paragraph under examination, and proposes a ready-to-use insertion formulation, complete with a reference to the page and source file. This is where the central architectural principle comes into play. The manuscript is not integrated into the corpus: it *queries* the corpus. Each paragraph of the manuscript is treated as a query sent to the index, which returns the most similar documentary passages based on semantic similarity. The historian’s text thus remains an active object, situated outside the established corpus, which prevents any contamination between the validated resources and the text under construction.

The pipeline proposes to characterize these relationships using a series of operational categories derived from practice: confirmation, contradiction, nuance, problematization, displacement, and particularization. These categories allow us to describe, in a simplified yet explicit manner,

the various ways in which a passage from the corpus can relate to a statement in the manuscript, bringing the historian's familiar practices closer to a manipulable ontology. Their construction follows a deliberately inductive logic: they were established based on recurring situations of confrontation between a statement and its materials, seeking a level of abstraction sufficient to be computationally formalizable, without sacrificing immediate recognition by the practitioner. The gesture corresponds to what McCarty (2005) refers to as the condition for truly heuristic modeling in the humanities.

Confirmation refers to cases where the segments used support the statement in question and converge with it. Contradiction corresponds to situations in which the available materials explicitly oppose the statement, whether factually or interpretatively. Nuance refers to intermediate cases, where the sources do not call the statement into question but limit its scope. Problematization refers to situations in which the corpus reveals that the statement corresponds to a question debated in historiography. Displacement refers to cases where the materials used suggest a different analytical framework, without necessarily invalidating the initial statement, but by re-situating it within a different interpretive geography—different scales, different categories, different temporalities. Finally, particularization refers to situations where specific examples or cases illustrate the statement while highlighting the limits of generalization.

These relationships between the pivot text and the corpus of associated texts are suggested to the user, who can also view them. Their mention serves as a signal that the practitioner is free to ignore or take seriously. For each analyzed paragraph, the generated report indicates the dominant profile, a document density score (corresponding to the number of passages semantically close to the paragraph of the text submitted to the LLM), the usable passages along with their classification and relevance score, the gaps identified in the document coverage, and a summary in a few sentences to guide the revision. Thus, when applied to the opening paragraph of a manuscript on the history of extradition in the 19th century, the system returns a high document density score, two passages that *support* the argument by offering a precise, sourced legal definition and a reflection on the paradoxical nature of the procedure, which precariously balances sovereignty and international cooperation, and one passage that *qualifies* the assertion regarding the long history of the practice by noting that the term “extradition” itself is not attested in an official French document until the decree of February 19, 1791. This information calls for a more cautious formulation regarding the historical continuity of the institution and points to the particular nature of a very ancient practice that was not legally defined until much later.

Applied at the manuscript level, this analytical framework enables a differential reading of the text, where areas of strong confirmation coexist with areas of contradiction, or of the new text being underdetermined by the existing one, and where one can track, section by section, the variations in the relationship to the parent texts. This type of mapping paves the way for what might be described, by analogy, as *computational criticism*, providing a framework in which the relationships between a text and its interlocutors are explicitly defined, comparable, and open to discussion.

Reading a historical argument

The second level of analysis is of a different nature. It is less a matter of asking “did someone say something that should be taken into account?” than “how, based on what, and according to what rules of transition, does your text come to say what it says?” Whereas critical mapping operates at the paragraph level, segment analysis subjects longer, coherent textual units—an argumentative sequence, delimited by the historian themselves—to modes of analysis more

deeply rooted in scholarly theorization. Three frameworks are employed. The first, drawn from the analytical philosophy of argumentation, breaks down reasoning according to Toulmin's (1958) model: thesis, data, law of transition, support, nuances, and anticipated refutations are identified and evaluated separately, making it possible to pinpoint missing components. This analysis is supplemented by a modeling approach based on Walton (1996), which allows us to classify the types of reasoning at work—causality, appeal to authority, analogy, argument from consequence—and to assess their robustness. In particular, it helps identify forms of argumentative fragility: hasty generalizations, excessive reliance on a single type of source, implicit use of questionable frameworks such as overly loose analogies or ill-founded appeals to authority, or failure to consider counterarguments.

The second framework is that of the New Rhetoric developed by Perelman and Olbrechts-Tyteca (Perelman & Olbrechts-Tyteca, 1958), which makes it possible to characterize the persuasion techniques at work in the text, the values mobilized, the type of audience implicitly targeted, and the risks of sophistical slippage. The writing of history, in fact, does not consist solely of articulating data, references, and interpretations according to logical rules; it also involves a process of discursive shaping, oriented toward an audience that must be convinced, at least temporarily, to accept a certain way of viewing the past. The choices of presentation, the references invoked, the analogies proposed, or the distinctions introduced all contribute to this dimension, which falls under what rhetorical theories have described as the conditions for acceptance.

RST (Rhetorical Structure Theory) complements this approach (Mann & Thompson, 1988) by exploring the relationships between discursive entities. Whereas New Rhetoric characterizes overarching persuasive strategies and the construction of the audience, RST analyzes how sentences and paragraphs are articulated to produce coherence or reveal, through their orphaned units and breaks in continuity, the missing rhetorical joints. A text may thus present a coherent persuasive strategy from a Perelmanian perspective while weakly articulating its argumentative units, or vice versa. This complementarity of scale is one of the reasons why the two frameworks are employed together rather than separately.

These analyses distinguish between logical validity and persuasive effectiveness. A line of reasoning may appear solidly constructed from a formal standpoint while relying on rhetorical devices aimed at reinforcing its acceptability; conversely, a weaker argument can be made convincing by the strength of its discursive framing, a carefully crafted introduction, a particularly evocative opening anecdote, a forceful use of analogy, or a metaphor rooted in cultural tradition. The approach does not aim to disqualify these operations, which are constitutive of all historical writing, but to make them visible and analyzable by situating them within a set of possible choices. It allows, for example, for the highlighting of forms of discursive caution, where the modalities, precautions, and explicit limitations reflect a concern for controlling generalizations—a concern that is sometimes obscured when the text is discussed solely in terms of its thesis and the arguments associated with it.

In the everyday practice of historical writing, these relationships and operations are rarely fully made explicit. The connections between empirical materials and interpretations often rest on implicit rules of transition, internalized through disciplinary training and shared within a scholarly community—what Bourdieu (1980) refers to as the disciplinary habitus, or Polanyi as tacit knowledge (Polanyi 1966): dispositions to act and reason that are objectively regulated without being the product of conscious obedience to explicit rules, sometimes to the point of becoming difficult to articulate even for the practitioners themselves. The proposed framework aims precisely to make these operations visible

The intended use of the pipeline is to map the text as a whole in order to identify areas worth closer examination, and then, where needed, to submit a complete argumentative segment — since the models used perform only moderately well at the paragraph level — corresponding to the zones flagged as worthy of attention by the overall mapping. The pipeline is thus best understood as a weak-signal instrument: designed not to diagnose, but to focus attention.

See the historian's work at the writing desk

These three frameworks can be applied separately or combined in a cross-analysis. It is this cross-analysis that yields the most actionable insights. When subjected to Toulmin analysis, a section of the same manuscript on extradition—focusing on how the Lamirande case was covered by the French press in 1866—reveals a generally solid argumentative structure: thesis, data, transition, and support are all present and identified. The data cited are precise: 259 press articles over a few months, close correspondence with French authorities, and abundant documentation. However, the transitional link between these data and the thesis—which concerns the narrative transformation of these accounts in the press—remains partially implicit. Perelman's rhetorical analysis, for its part, identifies a movement from factual observation toward a valorizing critical dissociation, with a clearly disciplinary target audience, as the author invokes the values of scientific rigor and historicity, with no detected use of sophistry. The analysis of rhetorical structures, for its part, indicates that the relationships between sentences are predominantly high-confidence elaborative relationships, with the exception of a contrastive relationship between two sentences regarding the nature of media narratives, whose confidence is rated as medium, signaling a passage where the argumentation could be made more explicit. The cross-analysis of the three frameworks converges on the same diagnosis: the strength of this segment lies more in its referential grounding and historical precision than in the formal robustness of its argumentation, which translates into a targeted recommendation: to make explicit the relationship between the accumulated evidence and the analytical conclusion, and to clarify the underlying warrant to strengthen logical coherence.

In fact, the analysis highlights a passage that reveals a discernible tendency in the text not to spell out the steps leading from the material to the conclusion in the context of writing a relatively short text, reflecting a practice that is only half-conscious: not spelling out the steps of a line of reasoning when only a limited number of characters are available, because this takes up a lot of space and it is doubtful that the publisher would be willing to negotiate an extension.

The combination of these approaches thus allows for a nuanced analysis of the texts. One can imagine it as the simultaneous use of several proofreaders, chosen precisely because they are likely to develop entirely distinct readings: one attentive to the text's grounding in literature, another to the strength of the argumentative sequence, and a third to rhetorical structure and discursive coherence—who do not offer a verdict, much less a measure, but rather advice, which, like all advice, can be ignored.

Development prospects: toward a situated analysis of historiographical practices

The pipeline presented here constitutes an initial formalization, still partial and deliberately designed as a professional prototype for writing assistance rather than as a platform. As such, it should be understood as an evolving tool, part of an exploratory approach, with numerous possible developments. This is not a finished model, but a working framework intended to be discussed, adjusted, and, if necessary, challenged by its users.

Argumentative graphs and visualization

A first extension concerns the structuring of identified relationships in the form of argumentative graphs. By representing statements as nodes and relationships of confirmation, contradiction, or nuance as edges, it becomes possible to produce visualizations of reasoning at different scales—paragraph, chapter, or even an entire work. These representations would make it easier to identify areas of argumentative density, points of tension, or critical dependencies between propositions, offering the historian an overview of structures that are often perceived only fragmentarily in linear reading. This approach aligns with work on argument visualization developed in the field of computational *argument mining* (Lawrence & Reed, 2020), whose graphical approaches could be adapted to the context of history.

Toward assisted annotation tools

A second area of development—which is a possibility rather than a promise at this stage—concerns the potential for semi-assisted annotation. The analyses produced by the pipeline—relational classifications, Toulmin components, Perelman techniques, and RST relations—constitute pre-annotations that trained historians could review, validate, or correct, following the human-machine loop annotation model now well-established in the field of *argument mining* (Lindahl, 2024). Such annotated corpora, built from the pipeline’s analyses and corrected by experts, would constitute valuable resources for the empirical study of argumentation in the historical sciences—and could, in the long term, inform supervised learning approaches capable of offering more refined diagnoses.

This prospect, however, raises important questions. Converting signals intended for historians at work into training data for automated models requires careful consideration of the conditions for the validity and transferability of these annotations across diverse historiographical contexts, as well as the role of human judgment in their interpretation. Signals designed to guide a revision may end up being treated as quasi-measures if they feed into a model without sufficient caution—a risk well identified in the literature on the computational annotation of argumentation (Lawrence & Reed, 2020; Lindahl, 2024).

Comparative Analysis of Argumentative Styles

A third extension concerns the comparative analysis of argumentative practices. By applying the framework to differentiated corpora—based on periods, geographical areas, or historiographical traditions—it becomes possible to compare the ways in which sources and bibliographies are utilized, arguments are structured, or certain rhetorical strategies are employed. Such an approach would pave the way for an empirical analysis of styles of reasoning in history, echoing Crombie’s (1994) work on styles of scientific thought and their philosophical reformulation by Hacking (1992), but this time anchoring the analysis in systematic descriptions of historical texts rather than in very general categories.

At this stage, the pipeline deliberately produces qualitative signals suggesting human intervention and verification rather than directly usable metrics. A quantitative extension is not ruled out, but it would require first measuring the stability of the core values obtained from established text corpora to determine whether they can effectively distinguish argumentative styles—a potential research direction that has not yet been implemented.

These perspectives can be linked to already well-established traditions. Linking argumentative structures with lexical-metric analyses would allow for the exploration of connections between

forms of reasoning, discursive configurations, and thesauri on a larger scale. Similarly, an integration with prosopographic approaches would open the possibility of linking argumentative styles to trajectories, institutional positions, or disciplinary affiliations, by situating texts within the social contexts of their production. We could thus envisage, in the long term, a social and intellectual history of argumentative practices, based on explicitly modeled indices rather than on impressions derived from reading—which, while sometimes informed, offer only fragile guarantees of generalizability.

Immediate applications: manuscript evaluation and writing instruction

The developments outlined above sketch out a potential medium- and long-term research program. However, we must not overlook the most immediate applications of the system, which require no further development and address documented needs. These stem from the system's roots in professional practice across its various aspects, as the tool's development has never been separated from the goal of supporting professional practice.

The first is the evaluation of manuscripts, whether they are dissertations, theses, or articles submitted to journals. Editors of academic journals face a growing challenge: finding available and competent researchers to evaluate the manuscripts submitted to them. This phenomenon, referred to in the literature as “*reviewer fatigue*,” results in review times that now exceed 100 days on average, with some journals taking more than six months to reach a decision, having direct consequences on the career trajectories of early-career researchers. In this context, the pipeline does not replace the disciplinary judgment of an expert reviewer—whose subject-matter expertise and engagement in ongoing scholarly debates remain irreplaceable—but it can produce, prior to or in addition to that judgment, a structured mapping of a text's argumentative strength, its positioning within historiography, and its rhetorical strategies. It thus offers an initial level of formalized analysis that can guide the human reviewer's reading or allow the author to strengthen their manuscript before submission.

The second immediate use is pedagogical. The teaching of academic writing in history—and more broadly in the humanities—most often relies on informal forms of mentorship: the thesis advisor who provides annotations, seminar participants who discuss a text's first draft, the colleague who proofreads. These practices are valuable but not equally accessible. Recent studies show that feedback-based tasks generated by LLMs on students' argumentative writing can be effective, particularly when they are part of an explicit pedagogical framework rather than unguided use—LLMs can play several roles in the development of argumentative skills, notably by providing evaluative feedback or by engaging as a collaborative work partner (Su et al., 2023). When applied to texts produced by master's or doctoral students, the pipeline offers a mode of analysis that goes beyond stylistic correction: it makes a text's argumentative structure visible, identifies it in relation to a historiographical corpus provided by the instructor, and allows the student to observe, in their own text, the categories they are otherwise expected to master.

The third use is the analytical reading of canonical texts in the field. Submitting a classic article in the discipline—a text by Furet, Ginzburg, or Davis—to the pipeline's analysis is not intended to produce an evaluation, but to shed light on its construction: according to which argumentative schemes does it organize its demonstration? which rhetorical techniques ensure its reception within the disciplinary community? This type of structured analysis of an established text constitutes an educational exercise in itself, revealing writing processes that ordinary reading

most often leaves implicit. In this sense, it aligns with the recommendations of a well-established reflexive tradition in the pedagogy of academic writing, which emphasizes the need to make explicit the disciplinary conventions that students are expected to internalize (Lea & Street, 1998).

Localization and Cultural Adaptations

One aspect rarely discussed in the literature on tools for analyzing academic texts deserves mention: the system is designed and calibrated for texts in French, within a European historiographical tradition. Extending it to other languages and cultural traditions would require substantial adaptation work, likely not limited to linguistic adjustments. Standards of historical writing, citation conventions, modes of engagement with historiography, and acceptable rhetorical registers vary considerably across national and disciplinary traditions. A system calibrated to the French tradition may produce biased results when applied to texts from different argumentative traditions—German-speaking, Latin American, or Asian historiography, to name just a few. This limitation is inherent to the project at this stage, and overcoming it would require interdisciplinary and intercultural collaboration far beyond the scope of the present work.

Interactive Interfaces and Accessibility

The development of interactive interfaces is another key challenge. Integrating the pipeline into writing or reading environments could enable real-time critical assistance, flagging inconsistencies with the corpus, weak arguments, or insufficiently substantiated generalizations. Such a use would bring the system closer to standard proofreading practices, which should be supported without turning them into automated procedures, leaving it up to the historian to decide whether or not to follow the suggested alerts.

To date, the pipeline requires the ability to work in console mode, sufficient proficiency in Python to set up a suitable environment and run a script, and access to a machine powerful enough to run local LLM models under acceptable conditions. We are aware that these conditions are not always met, and that an as precise and detailed as possible documentation—some might say verbose and rambling—is not enough to remove these barriers to entry. Extending this to other disciplines of scholarly discourse production, whose argumentative standards are not so far removed from those of historians, would moreover require adaptation and formalization efforts far beyond the objectives and expertise mobilized within the scope of this work.

Structural Limitations

These perspectives should not obscure the limitations of the framework, both in its current state and in its structural characteristics. The quality of the analyses depends closely on the corpus used, the segmentation choices, and the parameters of the models employed. The proposed categories remain simplistic and cannot capture the full complexity of historical reasoning. Finally, the risk of reifying the results—treating what is intended only as signals as verdicts—calls for maintaining a constant critical distance from the outputs produced.

The challenge is therefore not to substitute computational procedures for the historian's judgment, but to develop tools that can elucidate certain dimensions of it. The pipeline can thus be viewed as a tool for reflective exploration, intended to support practice rather than

standardize it, and to open a space for discussion on what historians actually do when they write, read, and evaluate texts.

Conclusion

The system presented in this article is not merely a technical tool designed to improve corpus exploration or assist in writing. It offers an initial instrumental formalization of the operations through which historical reasoning is constructed, justified, and discussed, for use by the historian at work, as well as by the historian of historical practices or the epistemologist, and perhaps others as well, since one can never be certain of the uses of the tools one puts into circulation. By linking the analysis of the relationship to the bibliography, the argumentative structure, and rhetorical strategies, it aims to make visible dimensions of the historian's work that most often remain implicit, and which are typically grasped through experience, mentorship, or revision, rather than through explicit descriptions.

This shift leads us to view architectures derived from language models no longer as mere devices for text production, but as environments for analysis and for testing reasoning. Generation no longer appears as an end in itself, but as a moment in a broader process of critical examination, where statements are situated, compared, and discussed in relation to identifiable historiographic corpora and traditions. AI is not here something to be avoided or endured, but an entity to be activated—provided it is activated by encoding professional practices into it, rather than allowing the implicit, and often opaque, norms that govern interaction with commercial platforms to be imposed.

The scope of this work is intentionally limited. It does not aim to provide an exhaustive model of historical reasoning, nor to replace disciplinary judgment with computational evaluation, nor to serve as a textbook generator. The pipeline remains exploratory, dependent on the corpora used and the formalization choices made. Its results should be interpreted as indicators, not as verdicts, and as invitations to reread, nuance, or reframe rather than as prescriptions for rewriting, even if one can never be certain that users will always follow the instructions.

This work, however, opens up a broader perspective. By making long-established operations analyzable and comparable, it helps shift the focus from the text to the practices that produce it. It thus suggests the possibility of instrumentalizing historiographical criticism itself, understood not as a set of explicit rules, but as a set of gestures, choices, and judgments embedded in writing, capable of being identified, described, discussed, and, to a certain extent, compared.

From this perspective, the figure of the augmented historian does not refer to a delegation of work to automated systems, but to an extension of the capacities for reflexivity regarding the conditions of production of historical knowledge, which explicitly codes the historian's role as an instrumented holder of expert judgment. The challenge is not to automate history, but to make the forms of reasoning that constitute it more visible, more open to discussion, and perhaps more comparable—by treating computational tools not as a substitute, but as a companion to historical practice, the mastery of which is necessary (Rygiel 2017), because failing to be able to influence the choice of techniques and the techniques themselves amounts to delegating choices that we believe we have shown to be as epistemic as they are political.

It is futile to hope to limit or prohibit the use of LLMs by professionals and students alike, if only because seeking to avoid the use of tools that enable significant productivity gains in a competitive social environment can only lead to the marginalization of those who virtuously

abstain, unless they are protected by their status. It may, however, be possible to develop—in dialogue with those whose profession is to design, code, and model—uses that foster practitioners’ reflexivity rather than resigning ourselves to delegating the generation of their texts—that is, of their thought—to an opaque system. Here we align with certain perspectives outlined by Jo Guldi, who advocates, in certain contexts, for a reflexive encoding of the historian’s practice (Guldi, 2014).

It remains to be clarified in what spirit this partnership is proposed here. The code produced as part of this work has something of a specification or a proposal about it. It explicitly formulates, in a debatable and modifiable form, the formalization choices made regarding professional practices. As such, it calls less for passive adoption than for dialogue: with historians who might wish to test, amend, or challenge these choices based on their own practices; with developers who could extend them in other directions; and with epistemologists who might find in these formalizations a subject for discussion. It is in this sense that the intended appropriation is neither passive nor client-like: it presupposes a reflexive interaction with the framework, which includes the possibility of questioning it down to its very foundations.

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Appendices

Extending the history

Examples of Pipeline Outputs — Appendix

This appendix presents three examples of actual pipeline outputs, as generated while working on a manuscript concerning the history of extradition and its media representations in the 19th century. Each block corresponds to a distinct level of analysis within the system.

Block	Level of analysis	Example
A	Assisted generation	Summary based on the corpus
B	Critical mapping (batch mode / RAG)	Section 1 of the manuscript — density score and qualified sources
C	In-depth segment analysis (cross-synthesis)	The Lamirande Case — Toulmin / Perelman / RST

Block	Level of analysis	Example
D	Tracking signals (batch mode / CPT)	§18 and §46 — codes and recommendations

In each case, the raw output is presented as is, followed by a brief comment indicating its effect on the manuscript—whether the signal was followed, qualified, or ignored, and why.

These examples are representative without being definitive. They were chosen because they clearly illustrate the three levels of the system and because they show both what the pipeline does well and the limitations that must be accepted.

Block A — Assisted Generation: Natural Language Query on a Corpus

This example illustrates the RAG function in its most direct use: the historian asks a question in natural language, the pipeline queries the corpus, selects relevant passages, and produces a written text in which each statement is anchored to an identified and localized source. The prompt explicitly requires precision, dating, and exclusive anchoring within the corpus—without resorting to external knowledge.

Instruction submitted to the pipeline

“Write a text describing the evolution of the political offense in the context of extradition in the 19th and 20th centuries, provide specific examples, date the developments, cite the relevant segments of the corpus, and conclude with a summary of the evolution without relying on external knowledge.”

Sources consulted by the pipeline (15 passages)

The pipeline queried the vector index and selected 14 relevant passages spread across five sources in the corpus. The sources used are as follows:

Source	Selected passages	Representative pages
Recjurid132_3F — Legal journal, case law collection	5	pp. 2, 5, 6, 10
Treaty on Extradition	5	pp. 52, 120, 124, 125, 127
Saint-Aubin — Extradition and Extradition Law	3	pp. 29, 45, 421, 429, 433

Text generated by the pipeline (excerpt)

The pipeline generates a text written in natural language. Each factual statement is followed by a reference to the source file and the corresponding page, allowing for direct verification. Only the first paragraph is reproduced here.

The evolution of political offenses in the context of extradition during the 19th and 20th centuries illustrates a profound transformation in the relationship between state sovereignty, criminal law, and international politics. Initially, under the Ancien Régime, extradition primarily targeted crimes against sovereignty, notably the crime of lèse-majesté, considered an affront to the honor of the state. Thus, acts classified as political offenses were those that undermined the sovereign’s authority, and their punishment was driven by a logic of honor rather than by classical criminal logic. This conception is evident in ancient treaties, such as the one concluded in 1174 between Henry II of England and the King of Scotland for the reciprocal extradition of traitors, or the

extradition request made in 1413 by Charles VI to the King of England for Parisian troublemakers [Extradition and Extradition Law, Saint-Aubin, pp. 421–429].

However, this approach gradually evolved beginning with the Enlightenment, which introduced a new conception of extradition focused on a criminal rather than solely political purpose. This change was accompanied by a legal distinction between political offenses and common law offenses, with the term “political offense” becoming a legal expression designating a struggle waged in the name of a political ideal, deserving a certain degree of leniency in light of the values emerging from 1789. This distinction became clearer in the 19th century, particularly with the recognition by certain states, such as France and Belgium as early as 1841, of the exclusion of political offenses from the scope of extradition, while other countries, such as Denmark, Sweden, Prussia, Russia, or Austria, continued to extradite for crimes of lèse-majesté [recjurid132_3F.pdf, p. 5]. [...]

Query metadata

Template	gpt-4.1-mini (API mode)
Temperature	0.1
Candidate passages evaluated	14
Corpus queried	Historical legal treatises + academic articles (law journal <1914)
Task constraints	No use of knowledge external to the corpus

What the historian did with it

The generated passage was not directly incorporated into the manuscript—that is not its purpose. It served as working material: the references identified by the pipeline (notably the Treaty on Extradition) were verified in primary sources and incorporated into the historiographical section. The “no external knowledge” constraint was deliberately tested here to assess the corpus’s coherence: the pipeline produced a dated and reasoned text, indicating that the corpus covered the issue sufficiently well to not require expansion given the passage’s broad framing function.

This block illustrates a central feature of the system: the traceability of statements. Each assertion produced by the pipeline refers to a segment located within the corpus, allowing the historian to verify, challenge, or dismiss each element independently. The text produced is not an autonomous synthesis but a grounded proposition, subject to expert judgment.

Block B — Critical Mapping (batch mode)

Submitted paragraph (§1 of the manuscript)

“Extradition is a technical procedure. Commonplace and governed by thousands of bilateral treaties, it is usually carried out amid general indifference. But all it takes is a high-profile case to bring it back into the spotlight—Julian Assange’s long flight, sought by Washington in the United Kingdom, Carles Puigdemont’s fight against his extradition to Spain, the reopening of an extradition corridor between France and Dubai—for old questions to resurface regarding the obligations binding sovereign states, sovereignty over bodies crossing borders, and the ownership of fugitives.”

Output from the pipeline (batch mode)

Document density score	0.80 (high)
Dominant profile	Confirmation + nuance
Passages analyzed	3 (out of 20 candidates)

Available resources identified by the pipeline

Relationship	Source	Usage suggested by the pipeline
citation	recjurid132_3F.pdf, p. 6	Precise legal definition of extradition — may strengthen the opening of the paragraph by clarifying its legal nature
citation	Saint-Aubin, p. 55	Illustrates the sovereignty/interdependence paradox (“mutual insurance against crime”)
note	Saint-Aubin, p. 51	Tensions between sovereignty, asylum, and extradition — enriches the discussion on moral and political issues

Gaps identified by the pipeline

- Lack of in-depth analysis of historical bilateral treaties and their role in the formation of international society
- No examples of historical cases illustrating the evolution of the procedure
- No discussion of the current limits of state sovereignty in a globalized context

What the historian did

Suggestion partially addressed. The quote from Saint-Aubin on “mutual insurance against crime” was incorporated into the revised version of the paragraph, as a footnote, to anchor the sovereignty/interdependence paradox. The gaps identified regarding bilateral treaties were deemed outside the scope of the article and were disregarded.

Block C — In-depth analysis by segment (cross-analysis)

Segment submitted — press coverage of the Lamirande case

“As a result, the Lamirande case sparked extensive correspondence with French authorities before becoming a famous, well-documented case, the repercussions of which led to the publication of the Canadian authorities’ correspondence on the subject. The French press reported on each of these episodes, devoting 259 articles to them during the few months of 1866 that spanned his flight, his pursuit, and then his return and trial. This narration of the pursuit and police work [...] represents more of a shift in the narrative than a transformation of the lexicon, which seems to bear little trace of it.”

Model: gpt-4.1-mini | Temperature: 0.1 | Length: 2,857 characters | Question asked: What are the evidential weaknesses of the passage?

Convergence table of the three frameworks

Dimension	A — Toulmin/Walton	B — Perelman	C — RST
Argumentative strength	Robustness: 0.50	Profile: 0.50	Development relationships (high confidence)
Evidence construction	Burden of proof: 0.50	Audience engagement: 0.50	Well-structured argument

Dimension	A — Toulmin/Walton	B — Perelman	C — RST
Overall coherence	Warrant coherence: 0.50	Value coherence: 0.50	Core message: P1
Rhetorical risks	Sophism: none	Risk: 0.50, none detected	0 low-confidence relationships

Cross-analysis (excerpt)

The segment’s argument relies primarily on narrative accumulation and referential authority (press, legal experts, crime fiction) to establish its credibility. The author clearly separates his analytical voice from the reported discourse, which reinforces the polyphonic clarity. However, the explicit logical coherence between evidence and conclusions remains partial, with sometimes tenuous connections in the reasoning, which may undermine the argument’s persuasive power.

Pipeline Recommendations

- Strengthen the explicit link between the evidence (press articles, legal analyses) and the analytical conclusions (see RST)
- Clarify the underlying warrants to solidify logical coherence (see Toulmin)

What the historian did with it

Recommendation followed. The recommendation regarding the implicit transition rule led to the addition of a sentence clarifying the link between the 259 identified articles and the conclusion on the narrative shift. The signal regarding the risk of undifferentiated reported speech was considered already addressed by the text and ignored. The overall diagnosis—referential soundness but weak transition—corresponds to a recurring trend in the writing of short academic texts, constrained by character limits, where the steps of reasoning are compressed, and the reader is expected to infer what is not written based on their habitus.

Block D — Locating signals (batch mode / CPT)

The batch mode analyzes the manuscript paragraph by paragraph and produces short signals geared toward identification. Here are two representative examples, taken from the same passage.

§18 — Compound signal

Detected signals	A: rheto_without_evidence F: axiological_coherence
Robustness / Profile	0.50 (model 09) 0.73 (model 10)
Warrant / Strength	0.80 0.70
Risk of fallacy	0.90 (09) 0.10 (10) — inter-model divergence
Walton scheme	Authority
Perelman technique	pragmatic argument
Recommended scripts	Script A (Toulmin on segment) Script B (Perelman)

Interpretation of Signal A (rheto_without_evidence): weak logical robustness but strong rhetorical profile—the author argues effectively without providing evidence. Top candidate for a Toulmin analysis on a segment. Signal F (axiological_coherence): strong consistency of values, low risk of sophistry—rhetorically solid area. Positive signal.

Note on inter-model divergence: Model 09 flags a sophistry risk of 0.90, while Model 10 rates it at 0.10. Rather than being masked, this divergence is documented in the output and flagged to the user as an area of uncertainty to be resolved by expert judgment.

§46 — Composite signal with discourse coherence alert

Detected signals	A: <code>rheto_without_evidence</code> D: <code>09_silent_10_speaks</code> F: <code>axiological_coherence</code>
Robustness / Profile	0.50 (09) 0.75 (10) — divergence reported
Walton Framework	authority
Perelman technique	authority_argument
Recommended scripts	Script A (Toulmin) Script B (Perelman) Script C (RST)

Signal D (`09_silent_10_speaks`): Model 09 did not identify an argument (neutral score 0.50), but Model 10 detects active rhetoric. Interpretation suggested by the pipeline: reported speech, narrative conclusion, or bibliographic list. To be examined to confirm if this is intended. The RST script is recommended over Toulmin: the issue may be one of discursive coherence between sentences rather than argumentation.

What the historian did with it

Signal §18 followed: the passage was run through script A (Toulmin) on the expanded segment, confirming the weakness of the transition rule. Revision completed. Signal §46 partially followed: the inter-model divergence on signal D prompted a re-reading of the paragraph, which turned out to contain a long quotation that was not clearly introduced and could lead a hurried reader to a false attribution. The introductory sentence was reworded. The RST script was not run—the problem was visible upon reading once attention was drawn to it.

These three blocks illustrate the complete chain of the system: from the batch signal oriented toward identification (Block C), to critical mapping through comparison with the corpus (Block A), to in-depth segment-level analysis (Block B). In each case, the pipeline produces signals that the historian is free to follow, qualify, or ignore—as these comments explicitly document.

ⁱ With all my thanks to Inria for providing a framework that made this project possible, to my colleagues in the Semis team, to Eric Guichard — even though he still thinks Perl is better — for never being short of a friendly objection, to the organisers of that wonderful place that is IXXI for making it thinkable, and to all those who love science, whose example is as stimulating and nourishing as it is demanding.

ⁱⁱ The scripts, documentation, output examples, and demonstration corpus are freely available on GitHub: <https://github.com/Datashs/Git/tree/main/AugmentingHistorian>.